

LII > U.S. Code > Title 10 > Subtitle A > PART V > Subpart F > CHAPTER 327
> SUBCHAPTER I > **§ 4401**

Quick search by citation:

Title

Section

Go!

10 U.S. Code § 4401 - Requirement for modular open system approach in major defense acquisition programs; definitions

U.S. Code Notes

(a) MODULAR OPEN SYSTEM APPROACH REQUIREMENT.—

A major defense acquisition program that receives Milestone A or Milestone B approval after January 1, 2019, shall be designed and developed, to the maximum extent practicable, with a modular open system approach to enable incremental development and enhance competition, innovation, and interoperability. Other defense acquisition programs shall also be designed and

developed, to the maximum extent practicable, with a modular open system approach to enable incremental development and enhance competition, innovation, and interoperability.

(b) DISCLOSURE REQUIRED.—

Not later than one year after the date of the enactment of this subsection, the Secretary of Defense shall make publicly available any standards for implementation of the modular open system approaches for contracts, unless the service acquisition executive with respect to a specific contract submits to the Secretary a request to not disclose such standards and the Secretary approves such request.

(c) DEFINITIONS.—In this chapter:

(1) The term “modular open system approach” means, with respect to a major defense acquisition program, an integrated business and technical strategy that—

(A) employs a modular design that uses modular system interfaces between major systems, major system components and modular systems;

(B) is subjected to verification to ensure that relevant modular system interfaces—

(i) comply with, if available and suitable, widely supported and consensus-based standards; or

(ii) are delivered pursuant to the requirements established in subsection (a)(2)(B) of section 804 of the William M. (Mac) Thornberry National Defense Authorization Act for Fiscal Year 2021, including the delivery of—

(I) software-defined interface syntax and properties, specifically governing how values are validly passed and received between major subsystems and components, in machine-readable format;

(II) a machine-readable definition of the relationship between the delivered interface and existing common standards or interfaces available in Department interface repositories; and

(III) documentation with functional descriptions of software-defined interfaces, conveying semantic meaning of interface elements, such as the function of a given interface field;

(C) uses a system architecture that allows severable major system components and modular systems at the appropriate level to be incrementally added, removed, or replaced throughout the life cycle of a major system platform to afford opportunities for enhanced competition and innovation while yielding—

(i) significant cost savings or avoidance;

(ii) schedule reduction;

(iii) opportunities for technical upgrades;

(iv) increased interoperability, including system of systems interoperability and mission integration; or

(v) other benefits during the sustainment phase of a major weapon system; and

(D) complies with the technical data rights set forth in sections 3771 through 3775 of this title.

(2) The term “major system platform” means the highest level structure of a major weapon system that is not physically mounted or installed onto a higher level structure and on which a major system component can be physically mounted or installed.

(3) The term “major system component”—

(A) means a high level subsystem or assembly, including hardware, software, or an integrated assembly of both, that can be mounted or installed on a major system platform through modular system interfaces; and

(B) includes a subsystem or assembly that is likely to have additional capability requirements, is likely to change because of evolving technology or threat, is needed for interoperability, facilitates incremental deployment of capabilities, or is expected to be replaced by another major system component.

(4) The term “modular system interface” means a shared boundary between major systems, major system components, or modular systems, defined by various physical, logical, and functional characteristics, such as electrical, mechanical, fluidic, optical, radio frequency, data, networking, or software elements.

(5) The term “modular system” refers to a weapon system or weapon system component that—

(A) is able to execute without requiring coincident execution of other specific weapon systems or components;

(B) can communicate across component boundaries and through interfaces; and

(C) functions as a module that can be separated, recombined, and connected with other weapon systems or weapon system components in order to achieve various effects, missions, or capabilities.

(6) The term “program capability document” means, with respect to a major defense acquisition program, a document that specifies capability requirements for the program, such as a capability development document or a capability production document.

(7) The terms “program cost targets” and “fielding target” have the meanings provided in section 4271(a) of this title.

(8) The term “major defense acquisition program” has the meaning provided in section 4201 of this title.

(9) The term “major weapon system” has the meaning provided in section 3455(f) of this title.

(Added Pub. L. 114–328, div. A, title VIII, § 805(a)(1), Dec. 23, 2016, 130 Stat. 2252, § 2446a; renumbered § 4401 and amended Pub. L. 116–283, div. A, title VIII, § 804(b)(1), title XVIII, § 1851(b)(1), (2), Jan. 1, 2021, 134 Stat. 3737, 4272; Pub. L. 118–159, div. A, title VIII, § 819, Dec. 23, 2024, 138 Stat. 1982.)



U.S. Code Toolbox

[Law about... Articles from Wex](#)

[Table of Popular Names](#)

[Parallel Table of Authorities](#)

[How current is this?](#)

[Accessibility](#)
[About LII](#)
[Contact us](#)
[Advertise here](#)
[Help](#)
[Terms of use](#)
[Privacy](#)

